

Dataset:

Temperature (X)	Play_Tennis (Y)
20	0
22	0
25	0
27	1
30	1

- $X \rightarrow$ Temperature
- $Y \rightarrow 0 = \text{No}, 1 = \text{Yes}$

Write the logistic regression model and solve the below questions for the given dataset.

1. the linear component z .
2. the sigmoid function.
3. the likelihood function for logistic regression.
4. the log-likelihood function.
5. the probability expression π .
6. the gradient of the log-likelihood.
7. the parameter update equation
8. Compute probability when $z=0.5$.
9. If threshold = 0.5, predict class.
10. Given the actual and predicted value calculate

Actual	Predicted
0	0
0	0
0	1
1	1
1	1

- a. Accuracy
- b. Precision
- c. Recall
- d. F1 score